

Mask-Guided Cotton Boll Phenotyping Architecture

Pre/post-defoliation UAV imagery → detector-prompted lint masks → proxy 2.5D review → conservative trait summaries

A. Pre/post UAV imagery

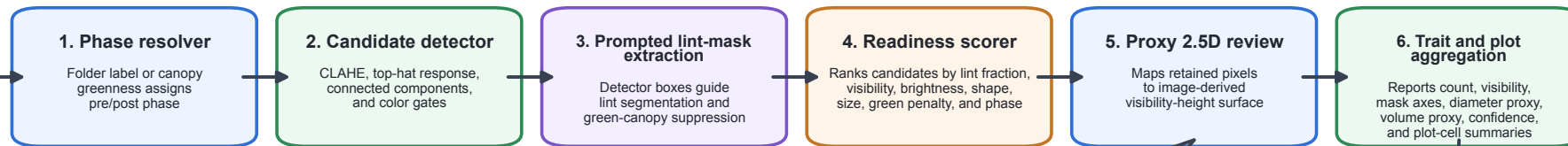
Pre-defoliation RGB



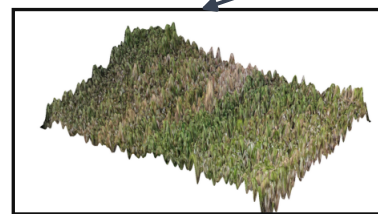
Post-defoliation RGB



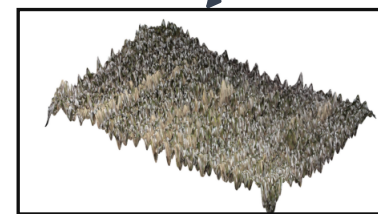
B. Implemented proxy pipeline



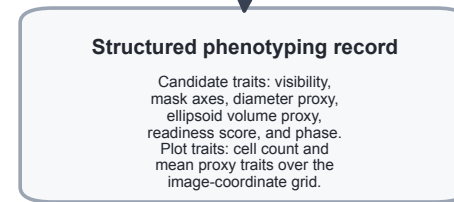
C. Proxy review and phenotyping outputs



Pre-defoliation proxy 2.5D



Post-defoliation proxy 2.5D



D. Calibration-dependent validation branch

